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19 December 1978
SUPERSEDING
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MILITARY SPECIFICATION

PLASTIC MOLDING MATERIAL, PRE-MIX, POLYESTER,

GLASS FIBER REINFORCED

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of glass fiber reinforced polyester pre-mix molding compound suitable for low pressure molding.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon

STANDARDS

FEDERAL
Fed. Std. No. 595 - Colors

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Army Materials and Mechanics Research Center, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

D 256 - Impact Resistance of Plastics and Electrical Insulating Materials
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing
D 638 - Tensile Properties of Plastics
D 695 - Compressive Properties of Rigid Plastics
D 790 - Flexural Properties of Plastics and Electrical Insulating Materials
D 792 - Specific Gravity and Density of Plastics by Displacement

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Material. The pre-mix molding compound shall consist of the following components:

(a) Low pressure polyester laminating resin
(b) Glass fibers, containing proper sizing and cut into suitable lengths to yield the properties required.
(c) Inorganic filler.
(d) Catalyst and promoter (if promoter is used).
(e) Inorganic pigments, as necessary, to give required color, when color is specified (see 3.4).

The components shall be proportioned, mixed and stored in accordance with suppliers' instructions (see 3.3 and 6.3). Due to a limited storage life, the pre-mix molding compound shall be used within the time prescribed (see 3.3). Reclaimed glass fibers and other materials are permitted when they meet all requirements specified herein and are essentially the same as the unused materials specified above for the pre-mix molding compound.

3.2 Property values. Specimens prepared from the pre-mix molding compound in accordance with suppliers' instructions shall conform to the property values specified in table I. Testing shall be in accordance with 4.2.4.



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Table I. Property requirements for molded specimens

| Property[1/] | Value required |
|---|---|
| Tensile strength, MPa (psi), min. | 93 (13,500) |
| Compressive strength, MPa (psi), min | 170 (25,000) |
| Flexural strength, MPa (psi), min. | 180 (26,000) |
| Impact strength, Izod, J/m (ft-lbs/in.) of notch, min. | 640 (12.0) |
| Barcol hardness, min. | 50 |
| Specific gravity, max. | 1.85 |

[1/] Metric equivalents: MPa = megapascal, J/m = joules per meter.

3.3 Instructions. The supplier shall furnish specifications for each of the components and complete instructions for the proportions, order and method of mixing the components, storage conditions including maximum shelf life under these conditions, and the method of molding test specimens and fabricated parts including temperatures, pressures, and cycle times.

3.4 Color. Unless otherwise specified, the color of the pre-mix material shall be the natural color of the material. When a specific color of molded material is required, the color and method of determining compliance shall be in accordance with Fed. Std. No. 595.

3.5 Workmanship. The prepared mixture shall be uniform in color and consistency, free from contaminants, and show no evidence of hardening, setting, or gelation. No scrap material shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection shall be performed in accordance with the provisions set forthe in MIL-STD-105 except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type, submitted for delivery at one time.



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4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of this specification and of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification will govern. A supplier's certificate of compliance with 3.1 shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 23 kilograms (50 pounds) for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2. The examination for instructions shall require only one complete set of instructions per lot.

4.2.2.1.1 Examination of the material for defects in color and workmanship. The sample unit for this examination, specified in table II shall be approximately one pound.

Table II. Examination for defects in instructions, color, and workmanship

| Examine | Defect |
|---|---|
| Instructions | Not furnished, or not in accordance with 3.3 |
| Color, appearance, and workmanship | Color of pre-mix material not as specified |
| | Color of molded material not as specified |
| | Color not uniform |
| | Consistency not uniform |
| | Not clean, presence of foreign material |
| | Evidence of hardening, setting, gelation or scrap material |



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4.2.2.1.2 Examination of the packaging. An examination shall be made in accordance with table III to determine that packing, and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Table III. Examination of packaging

| Examine | Defects |
|---|---|
| Packing | Not level specified; not in accordance with contract requirements |
| | Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability |
| | Inadequate application of components such as: incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers |
| Quantity of material | Less than specified or indicated quantity |
| Weight | Gross weight exceeds specified requirements |
| Markings | Interior or exterior markings omitted, illegible, incorrect, incomplete or not in accordance with contract requirements. |

4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

| Examination Paragraph | Inspection Level | AQL |
|---|---|---|
| 4.2.2.1.1 | II | 2.5 |
| 4.2.2.1.2 | S-2 | 2.5 |



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4.2.3 Classification of tests. All tests under this specification shall be classified as lot acceptance tests. They shall be made on each lot of material offered against this specification and, in conjunction with the above examination, shall be the basis of acceptance or rejection of the lot.

4.2.4 Testing. The material shall be tested for the characteristics listed in table I in accordance with the test method specified in table IV for each lot submitted for inspection. Sampling shall be in accordance with MIL-STD-105, and the lot size for the purpose of determining sample size for testing shall be expressed in units of 22.7 kilograms (50 pounds) of material. The inspection level shall be S-1. The sample size (number of units of product drawn from a lot) shall be in accordance with the single sampling plan for normal inspection and each unit shall consist of sufficient material to prepare all specimens required for testing. Each sample unit shall be tested and must meet all applicable requirements of section 3. Test results for each characteristic in each sample unit shall be the averaged results from the specimens tested for that characteristic.

4.3 Test methods.

4.3.1 Specimen preparation. Test specimens shall be prepared by molding in accordance with supplier's instructions (see 3.3).

4.3.2 Specimens conditioning and testing. Specimens shall be conditioned in accordance with procedure A of ASTM D 618 at a temperature of 23° ± 2°C (73.4° ± 3.6°F). Testing shall be at 23°± 2°C (73.4° ± 3.6°F) and 50 ± 5 percent relative humidity.



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Table IV. Number of specimens, dimensions, and applicable test methods

| Test | Specimen dimensions | Number of tests, min. | Test Method (ASTM) |
|---|---|---|---|
| Tensile strength | Type I, 3.175 millimetre (1/8-inch thick) | 5 1/ | D 638 |
| Compressive strength | As specified in the test method | 5 | D 695 |
| Flexural strength | 127 by 12.7 by 6.35 millimetre (5 by 1/2 by 1/4-inch) | 5 | D 790 (procedure A) |
| Impact strength, Izod | 6.35 millimetre (1/4-inch wide) | 5 | D 256 (procedure A) |
| Barcol hardness 2/ | Tensile strength specimen | 5 | see 2/ |
| Specific gravity | A portion of any of the above | 3 | Method A-1 of D 792 |

1/ Extra specimens should be prepared, as required for Barcol hardness determination.

2/ Barcol hardness shall be determined from direct readings taken with a Barcol Impressor. The average of a minimum of five readings taken on one tensile specimen shall be reported. The Impressor shall be used in accordance with the manufacturer's instructions. Readings shall be taken within 24 hours after specimen has been removed from mold.



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## 5. PACKAGING

Application. The requirements of section 5 apply only to purchase to or direct shipment to the Government.

5.1 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A in quantities of 90 kilograms (200 pounds), maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 180 kilograms (400 pounds), maximum.

Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.10 millimetre (0.004 inch) thick polyethylene liner properly heat sealed.

5.1.2 Level B. Unless otherwise specified, the material shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 90 kilograms (200 pounds) maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 180 kilograms (400 pounds) maximum.

Insofar as practical, drums shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with applicable container specification. Fiber drums shall be furnished with a 0.10 millimetre (0.004 inch) thick polyethylene liner properly heat sealed.

5.1.3 Commercial. Packing shall be in accordance with MIL-STD-1188.

5.2 Marking. In addition to any special marking specified in the contract or order, containers shall be marked in accordance with MIL-STD-129.

## 6. NOTES

6.1 Intended use. The pre-mix molding compound is intended for use in molded structures composed of the pre-mix molding compound alone or in conjunction with preimpregnated glass roving, glass cloth, or glass mat laminates where design requirements necessitate reinforced areas. The pre-mix molding compound is suitable for use in gunstocks and similar applications.



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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Instructions, and component specifications (see 3.3).
(c) Color, and method of determining compliance, as required (see 3.4).
(d) Selection of applicable level of packing (see 5.1).

6.3 Supplier. Unless material is forwarded directly to the Government the term supplier will be construed to mean the contractor who prepares and uses the pre-mix molding compound in the fabrication of molded items under direct contract with the Government or who subcontracts the fabrication of such items including the preparation of the pre-mix.

Custodian:
Army - MR

Preparing activity:
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Review interest:
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Project No. 9330-A783



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced documents(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-P-43043C (AR) | 2. DOCUMENT DATE (YYYYMMDD)<br>19 DECEMBER 1978 |
|---|---|---|

DOCUMENT TITLE
PLASTIC MOLDING MATERIAL, PRE-MIX, POLYESTER, GLASS FIBER REINFORCED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME *(Last, First, Middle Initial)* | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS *(Include Zip Code)* | d. TELEPHONE *(Include Area Code)*<br>(1) Commercial<br>(2) DSN *(if applicable)* | 7. DATE SUBMITTED *(YYYYMMDD)* |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE *(Include Area Code)* | |
|---|---|---|
| U.S. Army TACOM-ARDEC<br>Standardization Team | (1) Commercial<br>(973) 724-5822 | (2) DSN<br>880-5822 |
| c. ADDRESS *(Include Zip Code)*<br>Attn; AMSTA-AR-QAW-E<br>Picatinny Arsenal, NJ 07806-5000 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Standardization Program Office (DLSC-LM)<br>8725 John J. Kingman Road, Suite 2533<br>Fort Belvoir, Virginia 22060-6221<br>Telephone (703) 767-6888 DSN 427-6888 | |



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